



Clinical Practice Guidelines

Infusaport – Implantable Intravenous Delivery System

Scope

Site	Service/Department/Unit	Disciplines
SCGOPHCG	Organisation Wide	Nursing

Introduction

This guideline outlines the procedures and responsibilities of individuals and the health service organisation in relation to Infusaport- Implantable Intravenous Delivery System. It ensures the delivery of a uniform, integrated multidisciplinary approach with management that is consistent with best practice.

Risk Statement

Non-compliance with this guideline will:

Breach legislative requirements	☒	Impact on Patient Quality of Care	☒
Breach National/State/Hospital Policy	☒	Impact on Patient Safety	☒
Breach professional standards	☒	Misconduct	☐
Breach SCGH Mission & Values	☒	Staff Safety	☒
Other:	☐		

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These guidelines are intended to guide clinical decision making for clinical staff at North Metropolitan Health Service. They are not strict protocols and do not replace clinical judgement. These guidelines should never be relied on as a substitute for proper assessment of the particular circumstances of each case and the needs of each patient. If staff decide, in exercising their clinical judgment, not to follow the guideline, it is recommended that the reasons are clearly documented in the patient's health care record.

Principles

All personnel who insert and/or manage Infusaport or who prescribe or administer medications using these devices must adhere to the following:

- Medicines and Poisons Act 2014
- Medicines and Poisons Regulations 2016
- Pharmacy Act 2010.
- Work Health and Safety Act 2020

All personnel must observe the five moments of hand hygiene. Aseptic technique is to be maintained at all times during Infusaport insertion, management and removal.^{1,2} Follow aseptic technique and use the risk assessment outlined in the SCGOPHCG Infection Prevention and Management Aseptic Technique Policy.⁶

1.0 Insertion

1.1 Medical Requirements

Prior to procedure:

Referring medical team is responsible for:

- Obtaining a valid written consent (refer to WA Health Consent to Treatment Policy).
- Prescribing pre-operative 2% nasal mupirocin ointment and a chlorhexidine 2% or 4% body wash as per the Mupirocin Nasal Ointment for Decolonisation Guideline.
- Review recent blood results including coagulation profile.
- Insertion of a peripheral intravascular catheter (PIVC) prior to insertion of the Infusaport.

Procedure:

The only personnel designated to insert Infusaport are Interventional Radiologists and Surgeons.¹⁻³

Obtain chest X-ray to confirm correct catheter tip placement in the superior vena cava and exclude pneumothorax, except when Infusaport is inserted under fluoroscopy (X-ray guidance).³

Post Procedure:

- Catheter tip position should be documented in referral / operation report, e-Referral, or the digital medical record (DMR), along with the absence or presence of a pneumothorax and any other clinically relevant procedural information.^{4,5}
- Complete the decolonisation regime if the emergent need for an Infusaport did not permit completion prior to insertion. Once decolonisation regime is completed, ongoing use of mupirocin is not required⁶ refer to the SCGOPHCG Guideline – Mupirocin Nasal Ointment for Decolonisation for details.

1.2 Safety Information

Infusaport insertion must be performed in Theatre or Interventional Radiology by designated personnel using full surgical aseptic technique.^{1,2,4,5}

Assess the insertion site at least once a day for signs of infection or dislodgement. Document assessment in the digital medical record.

Patients with known Heparin-Induced Thrombocytopenia Syndrome (HITS):

- **DO NOT** flush with heparinised saline or lock with high dose heparin.
- Seek medical advice.⁷

All patients with an Infusaport insitu should have a device identity card or wristband. All clinicians should sight the patient's identity card or wristband prior to needling.

If the card or wristband is unavailable, the clinician needs to be aware of the type of Infusaport that is insitu prior to needle insertion, by contacting the IV Access Service for confirmation.

1.3 Nursing Management

Pre-Procedure:

Refer to the SCGOPHCG Guideline – Mupirocin Nasal Ointment for Decolonisation for details:

- Pre-procedure apply nasal mupirocin 2 % ointment AND application of chlorhexidine 2% or 4% body wash **for five days prior to the procedure.** ^{3,4}
- Patient to have a shower / bathe using pre-operative Chlorhexidine 4% body wash.⁶ provided by the hospital on the morning of the procedure. Administer any IV antibiotics, as prescribed.
- Patient must wear a hospital gown.
- Hair removal is **not recommended** unless it interferes with dressing adhesion.⁵
- Consent form, observation charts and preoperative checklist should accompany the patient.

Post- insertion of Infusaport:

For any emergency procedure where the full decolonisation regimen cannot be completed prior to the procedure, each therapy (mupirocin ointment and chlorhexidine wash) should be initiated as soon as possible and continued until completion of the full five-day decolonisation regimen.

- Provide the patient with an Infusaport patient information leaflet.
- Following Infusaport insertion, provide the patient with an identity card and wrist band containing Infusaport details specific to the type of device.
- Instruct patient to carry the identity card and wear wrist band at all times.
- If Infusaport is **not needed** post insertion, a Tegaderm CHG™ dressing will be placed over the insertion site, and the patient will be instructed to leave the dressing insitu for up to 7 days. Patient to be given 2 x spare Tegaderm +Pad™ dressings to take home.
- If Infusaport is **needed** post insertion, place an antimicrobial impregnated dressing e.g. Biopatch® or Infusaport Tegaderm CHG under the needle, cover with a sterile occlusive transparent dressing (e.g. Opsite IV 3000) as required, and document this on the Infusaport Nursing Care Plan and in the digital medical record. The Infusaport needle can remain insitu for up to 7 days.

If the patient has a known Chlorhexidine allergy use an alternate antimicrobial agent e.g. Kendall™ disc.

- Neck puncture site will be covered with Steri Strips™ and a Tegaderm +Pad™ dressing, and patient given 2 x spare dressings to take home. Instruct patient to keep the site covered and dry until the wound has closed over.
- Educate the patient on signs of infection to monitor for at both wound sites (erythema, swelling, heat, pain, purulent discharge, and fever). Provide the patient with instructions on what to do if they suspect infection.
- Confirm the patient has timely follow up with their treating facility and if not, please arrange prior to discharge.

Device Information

- An Infusaport is either surgically inserted or via Radiological guidance, and is a permanently implanted venous device consisting of:

- A port body (single or double septum / reservoir) with silicone membrane for accessing;⁷ and:
- A catheter (single or double) which is threaded into a large vessel, with optimal placement in the lower third of the Superior Vena Cava (SVC).⁷
- The majority of Infusaport inserted are a single lumen device, however a double lumen Infusaport may be considered based upon individual patient assessment and IV needs.
- The majority of Infusaport are inserted in the upper left or right chest via the internal jugular vein. Alternative sites are the subclavian or brachial veins.^{1, 7} The femoral vein may be considered as a last resort.

2.0 Needle Insertion

Ideally the Infusaport should not be accessed for fourteen days post insertion. If the Infusaport must be used within fourteen days due to clinical urgency the medical team must document this in the DMR.

2.1 Safety Information

Personnel authorised to insert a needle to access the Infusaport must have achieved competency in needling Infusaport, by completing the Infusaport competency assessment tool.^{1,3,7}

Seek immediate medical assistance if suspected:

- **Catheter rotation** - Inability to palpate the septum of the Infusaport may indicate that the Infusaport has flipped within the subcutaneous pocket housing.
- **Pinch-off syndrome** - Occurs when the Infusaport is inserted via the subclavian vein and becomes compressed between the clavicle and first rib.
- **Erosion** - Results from loss of viable tissue over the implantation site exposing the portal body.

Use only non-coring needles to access the Infusaport^{4,8,9}

Do not use standard hypodermic needles as these may damage the septum.

If a patient has a Power-Injectable Infusaport (PII) insitu and is requiring the injection of contrast for CT scan, the Power-Injectable Infusaport **MUST** be needled with a specific PII needle (obtained from Radiology, the IV Access Service, or G73).

At all other times, use an ordinary non-coring needle (e.g. Gripper or Safe Step™) for all Infusaport.

DO NOT use a syringe smaller than 10ml volume to access an Infusaport. Smaller syringes can increase pressure within the line and cause potential damage or rupture of the Infusaport.

In case of difficulty aspirating blood or flushing the Infusaport:

- Do not initiate injection or infusion therapy.
- Organise an expert nursing review via the IV Access Service to confirm Infusaport integrity.
- Consider repeat chest X-ray to confirm catheter tip position.
- Consider lineogram/portogram to exclude / confirm fibrin sheath occlusion.
- Injury or loss of device may be as a result of extravasation, infiltration, catheter embolism or infection.
- Contact the IV Access Service or After Hours CNS/CNM for advice.

2.2 Nursing Management

- Check insertion documentation to identify the type of Infusaport (open or closed ended) and if it is a power injectable Infusaport
- Consider applying Emla® cream to the insertion site, 30-60 mins prior to accessing the Infusaport, as needling can be painful.⁸
- The Infusaport permits repeated access to the vascular system for the delivery of all intravenous therapy including medications, parenteral fluids, blood products and for sampling of venous blood (intravenous ports only).¹
- Prior to accessing the Infusaport inspect the insertion site for signs of infection or malposition. If infection at the Infusaport site is suspected escalate to a medical review. Signs to monitor for, include^{1,4,5,8}
 - Swelling, redness, pain, ooze/exudate
 - Haematoma, accumulation of serous fluid
 - Device rotation or extrusion.
- During administration: Observe insertion site for signs of swelling. Ask the patient to report immediately any burning, pain or discomfort.^{1,4,5}
- If any of these symptoms are observed or experienced immediately cease infusion, seek medical review and document.
- Assess insertion site daily while the Infusaport is needled.
- The Infusaport needle needs to be changed every 7 days, when in use.^{4,9}
- Direct problems/queries to:
 - IV Access Service,
 - Clinical Nurse Specialist - Oncology/Haematology,
 - Medical team,
 - After Hours CNS/CNM.

2.3 Procedural Information

Note: * Double quantity required if accessing both lumens of a double septum Infusaport⁶

REQUIREMENTS

- Dressing trolley⁸
- Blue Disposable Underpad
- Waste Disposal bag⁸
- Dressing pack⁸
- Chlorhexidine 2% in 70% alcohol solution;^{5,8,9,10} (Patients with known allergy; consider alternative solution e.g. Povidone-iodine in 70% alcohol or Prontosan®).
- Personal Protective Equipment, including sterile gloves^{8,9}
- Non-coring Needle* (e.g. Gripper™, Safe Step™, or Powerloc™).^{4,8,9} (Determine correct needle length to access Infusaport by palpating the septum)
- 2 x 10 ml Sterile BD Posiflush™ Pre-Filled Saline Syringes*, **or** 2 x Sodium Chloride 0.9% ampoules and 2 x 10 ml Luer Lock syringes⁸
- 1 x 10 ml Luer Lock syringe (for blood discard) and extra if required for blood sampling.
- 1 or 2 x positive pressure valves (PPV), depending on the type of non-coring needle used.⁸
- Antimicrobial impregnated dressing e.g. Biopatch®, Kendall™ disc or Infusaport Tegaderm CHG.^{1,7,10,12}
- If not using an Infusaport Tegaderm CHG, a sterile occlusive transparent dressing (e.g. Opsite IV 3000™)^{8,12}
- Additional securing strips (e.g. Fixamol™ or Hyperfix®)
- Cavillon™ barrier prep

- Blood tubes required for sampling and a blood transfer device
- **ADDITIONAL REQUIREMENTS FOR “OPEN-ENDED” INFUSAPORTS:**
- Heparinised saline 50 units in 5mls (if not being used immediately) *⁸
- Heparin 5000 units in 5ml (if locking pre de-needling)
- 1 x 10ml Luer Lock syringe
- 1 x drawing up needle

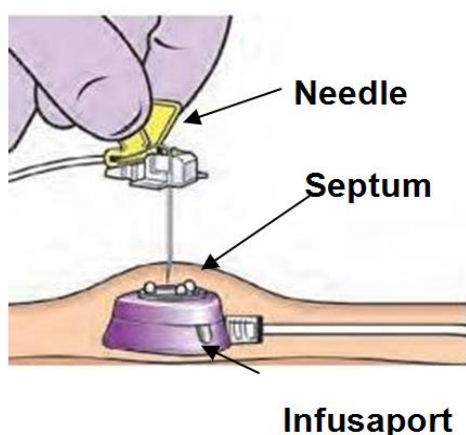


Diagram 1 Needling the Infusaport



Diagram 2 Location of Infusaport on Chest Wall

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Procedure

1. Confirm patient's identity, explain the procedure, and obtain verbal consent.^{8,9}
2. Perform hand hygiene.^{1,7,8,11}
3. Prepare and assess the patient.⁸
4. Inspect insertion site and palpate the Infusaport septum to determine needle length. Consider the application of Emla® at Infusaport site, as required.⁸
5. Perform hand hygiene.
6. Place blue disposable underpad across patient's upper chest, below site of insertion
7. Clean trolley with combined detergent/disinfectant wipes, as per hospital policy.⁸
8. Gather all equipment and select appropriate size and type of non-coring needle.
9. Perform hand hygiene.^{7,8,11}
10. Prepare the sterile field. Open dressing pack using the corners, add items, and set out equipment using aseptic non-touch technique.^{1,4-6}
11. Perform hand hygiene, and don sterile gloves.^{1,4-9,11}
12. Place sterile sheet on top of the blue disposable underpad and close to the Infusaport site.⁸
13. Clean the Infusaport site liberally with Chlorhexidine 2% in 70% alcohol or an alternate non-Chlorhexidine based solution e.g. Prontosan® for patients with a Chlorhexidine allergy. Clean in a back-and-forth to a radius of at least 5cm.^{1,4,7-9}
14. Allow to air dry.⁸
15. Prime tubing and non-coring needle with Sodium Chloride 0.9% and attach primed PPV on both access ports using non touch technique.
16. Slip the antimicrobial disc onto the upturned needle foam side up so it will make full contact with the patient's skin when the needle is inserted or apply an Infusaport Tegaderm CHG dressing to the dry skin over the Infusaport site prior to needling.

17. Once site completely dry, and/or post Tegaderm CHG dressing placement, palpate and anchor the Infusaport site between index and middle finger of non-dominant gloved hand.^{8,9}
18. When using the antimicrobial disc:
 - a. Using the dominant gloved hand, hold the antimicrobial disc and Infusaport needle together as you insert the needle at 90° into the centre of the Infusaport, avoiding any previous puncture sites, and using constant gentle pressure until it touches the base of the septum⁸- (refer to Diagram 1 and 2)
19. Use a 10ml syringe to withdraw 5mls of blood to verify correct needle placement. Discard blood (unless blood cultures required).^{8,10,13}

Note: Difficulty withdrawing blood may indicate catheter compression, obstruction, incorrect needle size or incorrect placement. If unable to obtain blood, do not remove needle. Seek assistance from the IV Access Service or senior nursing staff.
20. Flush with 20mls Sodium Chloride 0.9% using 2 x 10ml syringes using a rapid pulsating push/pause action.^{8,13} (Note: may require multiple flushes to clear blood from line and PPV)
21. Secure with a sterile occlusive transparent dressing (e.g. Opsite IV 3000™) using non-touch technique to prevent disconnection or dislodgment.¹² Further secure dressing with an adhesive non-woven fabric border. E.g. Fixomul™
22. When PPV are insitu on access ports, ensure the clamps are open.
23. Dispose of sharps and waste appropriately.
24. Remove gloves and perform hand hygiene.^{1,7,8,11}
25. Clean trolley with combined detergent/disinfectant wipes.
26. Perform hand hygiene.^{7,8,11}
27. Document insertion date, insertion site, needle size, type of port, and due date of next needle and dressing change on the Infusaport Nursing Care Plan and in the digital medical record.^{9,10}
28. Documentation must also include the patency of the Infusaport (blood withdrawal and flushing ability) and assessment of the Infusaport site and surrounding skin in the digital medical record.
29. If Infusaport needle is not remaining insitu, document due date of next access, flush and lock on the Infusaport Nursing Care Plan and in the digital medical record.

3.0 Flushing

3.1 Medical Requirements

Open-ended Infusaport only: Heparinised saline 50 units in 5mls flushing solution must be prescribed by a Medical Officer.

3.2 Safety Information

Identify the type of Infusaport insitu prior to administration.

Observe the five moments for hand hygiene. Follow aseptic technique and use the risk assessment outlined in the SCGOPHCG Infection Prevention and Control Aseptic Technique Policy. This must include the use of appropriate Personal Protective Equipment (PPE) to prevent contamination and mucosal or conjunctival splash injuries⁶.

DO NOT:

- use a syringe smaller than 10ml volume when flushing.
- force fluid into the Infusaport to avoid excess pressure and potential catheter rupture.

Note: A syringe of ≥10mL volume is used to prevent increased pressure and potential catheter rupture.^{4,5}

Seek assistance from the IV Access Service or senior nursing staff if there is a suspected Infusaport blockage.

3.3 Nursing Management

- Pre and post administration of IV therapy and post blood sampling: Flush with 20mls Sodium Chloride 0.9%. Be mindful of excessive flushing of Infusaport for patients on strict fluid restrictions.
- **Open-ended** Infusaport only: Follow with a flush of Heparinised saline 50 units in 5mls.

3.4 Procedural Information

Requirements

- Clean kidney dish or blue tray.
- 2 x 10ml BD Posiflush™ Pre-Filled Saline Syringes OR 2 x Sodium Chloride 0.9% for injection and 2 x 10 ml luer lock syringes
 - 2 x large 70% alcohol wipes
- PPE including non-sterile gloves^{6,8}
- **ADDITIONAL REQUIREMENTS FOR “OPEN-ENDED” INFUSAPORTS**
- 1 x 10ml Luer Lock syringe
- 1 x Heparinised saline 50 units in 5mls
- 1 x drawing up needle
- Sharps container

Procedure

1. Perform hand hygiene.^{1,7,8,11}
2. Put on non-sterile gloves.
3. Using a non-touch technique scrub the end of the PPV with a large 70% alcohol wipe for 30 seconds using friction and different areas of the wipe. Allow to air dry.
4. Flush Infusaport with 20 ml Sodium Chloride 0.9% using luer lock syringes, OR 2 x 10ml BD Posiflush™ Pre-Filled Saline Syringes; using a rapid, pulsating push/pause positive pressure technique. Disconnect the syringe before the flush volume empties (bottoming out) to reduce reflux of blood back into the lumen and catheter occlusion.^{8,13}
5. Dispose of all waste and any sharps, in sharps container provided.
6. Remove gloves and perform hand hygiene.^{7,11}
7. Observe the insertion site for signs of swelling. Ask the patient to report immediately any burning, pain or discomfort.⁹ If any of these symptoms are observed or experienced: Cease flushing, seek medical review and document.

Additional Information

1. Open-ended Infusaport: Follow with a flush of Heparinised saline 50 units in 5mls.

4.0 Locking and Needle Removal

4.1 Medical Requirements

Open-ended Infusaport: A medical prescription for locking solution of Heparin 5000 units in 5mls is required.

If an alternate catheter locking solution (e.g. Taurolock™) is required, this must be prescribed on the medication chart by an authorised prescriber.

4.2 Safety Information

Personnel must have achieved competency in de-needling Infusaport, by completing the Infusaport competency assessment tool.^{1,3,7}

Identify type of Infusaport insitu prior to administration.

DO NOT use a syringe smaller than 10ml volume to prevent increased pressure and potential catheter rupture.

4.3 Nursing Management

- Change Infusaport needle every 7 days or as required.⁹
- Flushing requirements prior to needle removal and locking:

Closed-ended Infusaport	• Prior to needle removal, flush with 20mls Sodium Chloride 0.9%, • Closed-ended infusaport <u>not in use</u>: flush with 20mls Sodium Chloride 0.9% every 4-6 weeks.
Open-ended Infusaport	• Prior to needle removal, lock with Heparin 5000 units in 5mls. Do not use Heparin lock more than once weekly (this Heparin dose is close to the therapeutic range). • Open-ended Infusaport <u>not in use</u>: Flush with 20mls Sodium Chloride 0.9% and lock with Heparin 5000 units in 5mls every 4-6 weeks.

For patients at high risk or with a history of catheter related blood stream infections, consider consultation with Infectious Diseases, in conjunction with the patient's specialist team, for the use of an alternative locking solution, (e.g. such as TauroLock™)

- TauroLock™ or Taurolidine and Sodium Citrate 4%, can be instilled into the Infusaport, up to a volume of 2mls, prior to needle removal.

4.4 Procedural Information

Requirements

- Dressing Trolley
- Waste Bag
- 1 x large 70% alcohol wipe
- 2 x 10ml BD Posiflush™ Pre-Filled Saline syringe; **OR** 2 x 10 ml Sodium Chloride 0.9% ampoule and 1 x 10ml Luer Lock syringe
- PPE including non-sterile gloves⁶
- 1x sterile pressure dot
- Sharps container
- 1 x drawing up needle

ADDITIONAL REQUIREMENTS FOR "OPEN-ENDED" INFUSAPORTS

- 1 x 10ml Luer Lock syringe
- Heparin 5000 units in 5mls
- 1 x drawing up needle

Procedure

1. Perform hand hygiene and put on non-sterile gloves.^{1,7,8,11}
2. Remove dressing and discard, observe insertion site for signs of infection.
3. Remove gloves and perform hand hygiene.^{1,7,8,11}
4. Put on non-sterile gloves.
5. Using a non-touch technique scrub the end of the PPV with a large 70% alcohol wipe for 30 seconds using friction and different areas of the wipe. Allow to air dry.¹⁰

6. Using a non-touch technique, flush Infusaport with $\pm$ 2 x 10ml BD Posiflush™ Pre-Filled Salines, or 2 x 10ml Sodium Chloride 0.9% ampoules, using a rapid, pulsating push/pause positive pressure non-touch technique.^{8,13} Disconnect the syringe before the flush volume empties (bottoming out) to reduce reflux of blood back into the lumen, decreasing the risk of catheter occlusion.^{4,5}
7. **Open-ended Infusaport:** Follow with Heparin 5000units in 5mls.
8. Stabilise the Infusaport by placing your gloved thumb and forefinger on either side of the Infusaport during needle removal.
9. Place finger on lip of non-coring safety needle (e.g. Gripper™, Safe Step) and lift up until audible 'click' is heard (needle retracts for safety).
10. Using a non-touch technique, place sterile pressure dot over exit site.
11. If Infusaport site is soiled post needle removal, clean Infusaport site with Chlorhexidine 2% in 70% alcohol solution or Prontosan® and allow to dry prior to placement of pressure dot.
12. Educate patient to keep dressing dry and remove pressure dot after 24 hours and monitor Infusaport site for any signs of infection (erythema, swelling, heat, pain, purulent discharge, and fever).
13. Instruct patient to seek medical advice from their treating facility (e.g. The Cancer Centre or the patient's General Practitioner) if they have concerns regarding infection at the Infusaport site.
14. Dispose of all waste and any sharps in sharps container provided.
15. Remove gloves and perform hand hygiene.^{1,7,8,11}
16. Document date of flush/lock and needle removal, and due date for next flush/lock in the digital medical record and on the Infusaport Catheter Nursing Care Plan. Ensure the patient has a required follow up booked, with either their treating facility or Silver Chain prior to discharge.

5.0 Blood Sampling

5.1 Medical Requirements

Medical Officer is required to complete a pathology request form.

5.2 Safety Information

An Infusaport is often inserted because a patient has poor peripheral access, making blood sampling from an Infusaport necessary.

Sampling blood from an Infusaport increases risk of catheter occlusion. Use a pink vacutainer and syringe, and flush immediately post sampling to reduce occlusion risk (as detailed in procedure).

Bloods may be obtained from an Infusaport **except** during infusion of inotropic drugs, TPN or Heparin. Do not cease these infusions for blood sampling, use an alternative route.

All specimen tubes require 3 points of identification and should be labelled immediately after collection.

- Specimens must be labelled by the collector at the bedside.
- Clearly label each specimen with the patient's full name, UMRN, date of birth, time and date of collection, and collector's signature. Use patient's hospital label when possible.
- Blood culture bottles also require information about the sampling site (i.e. peripheral, CVC, PICC, or Infusaport).
- Specimens without patient's full name and date of birth will be rejected by the laboratory.

5.3 Nursing Management

- Routine sampling increases the risk of catheter occlusion and infection. Ensure the catheter is flushed well post sampling and that all PPV are clear of blood following collection.^{4,5}
- To maintain a closed system and reduce the risk of infection and contamination **do not** remove the PPV from the end of the extension set on the Infusaport needle.^{1,4,5}

5.4 Procedural Information

Directly attach syringes to the PPV, after appropriate cleaning.

Requirements

- Dressing trolley or 2 x kidney dishes or a blue tray
- 4 x 10ml Luer Lock syringes
- 1 x 20ml Luer Lock syringe
- 2 x 10 ml Sodium Chloride 0.9% ampoule and 2 x 10 ml Luer Lock syringe, OR 2 x 10ml BD Posiflush™ Pre-Filled Saline Syringe
- 1 x large 70% alcohol wipe
- Vacutainer syringe adaptor
- Blood specimen tubes
- Sharps container
- PPE including non-sterile gloves^{6,7}

ADDITIONAL REQUIREMENTS FOR “OPEN-ENDED” INFUSAPORTS

- 1 x 10ml Luer Lock syringe
- Heparinised Saline 50units in 5mls
- 1 x drawing up needle

Procedure

1. Turn off infusions one minute prior to sampling if possible.
2. Collect all required equipment.
3. Confirm patient's identity, explain the procedure, and gain verbal consent.
4. Perform hand hygiene.^{1,7,8,11}
5. If used, clean trolley or blue tray with a combined detergent/disinfectant wipe.
6. Perform hand hygiene.^{1,7,8,11}
7. Prepare all equipment.
8. Perform hand hygiene and apply non-sterile gloves.^{1,7,8,11}
9. Clean end of positive pressure valve with a large 70% alcohol wipe, using a non-touch technique, scrub using friction and different areas of the wipe for 30 seconds and allow to air dry.¹⁰
10. Attach a 10mL Luer Lock syringe using non-touch technique and aspirate 5mLs of blood. Discard syringe and blood to prevent an abnormal reading from contaminated blood.
 - NOTE: If taking blood cultures, do not discard first 5mls. Refer to SCGH Infection Control Prevention and Control Manual/ Specimen Collection for Microbiological Examination Guideline.
11. Attach 20ml Luer Lock syringe, using a non-touch technique directly to Infusaport PPV, and withdraw blood sample. Then attach vacutainer syringe adaptor to the syringe, and transfer blood into the blood specimen tubes.
12. Using a non-touch technique, flush with 20 ml Sodium Chloride 0.9% using a Luer Lock syringe, **OR** 2x 10ml BD Posiflush™ Pre-Filled Saline Syringes. Use a rapid, pulsating push/pause positive pressure technique, to completely clear lumen of any visible blood. Additional flushing may be required.^{4,5,13}
13. Discard all waste and sharps appropriately.
14. Remove gloves and perform hand hygiene.^{1,7,8,11}

15. Label and sign specimens and send to Pathology.

6.0 Unblocking Lumens with Alteplase

6.1 Medical Requirements

Alteplase must be prescribed on the WA hospital medication chart by an authorised prescriber. This should be prescribed as a once off medication, specifying the dose, volume, device and intended lumen.

6.2 Safety Information

Alteplase is available as a 2mg in 2 mL prefilled syringe and is supplied by pharmacy. Alteplase is indicated for catheters that are resistive to flush or aspirate blood to promote patency of the central venous access device and to resolve clotting.^{12,13,14,15}

- Alteplase 2mg/2mL pre-filled syringes are stored frozen in pharmacy and issued for individual patient use
- Visually inspect the contents of the syringe to confirm it has completely defrosted prior to administration
- If not for immediate use, store in the fridge and use prior to the labelled expiry date and time

ONLY USE Alteplase FOR CATHETERS RESISTIVE TO FLUSH AND/OR ASPIRATE BLOOD.

Do not use Alteplase:

- if a catheter-associated infection is suspected.
- to unblock lumens being used for infusions containing inotropic drugs or vasoactive drugs. Lines with inotropic or vasoactive drugs are not routinely flushed hence use of Alteplase may unintentionally expose patients to inotropic or vasoactive drugs left in the line.
- if unable to aspirate / inject any fluid into the line, contact the IV Access Service for advice.

Potential complications of use include:

- Increased anticoagulant effect
- Haemorrhage; note any Alteplase released into the circulation is metabolised rapidly by the liver (plasma half-life = 5 mins) so risk of bleeding is minimal.
- Anaphylaxis (rare < 2%)
- Rapid lysis of coronary artery thrombi causing cardiac arrhythmias (MIMS online, 2008)

6.2 Nursing Management

Exclude other possible causes of occlusion i.e. kinked catheter, incorrect position, clamped tubing or needle dislodgement.

Try clearing the line with a 20ml Sodium Chloride 0.9% flush, using a vigorous, pulsating (stop-start) technique. If unable to inject any fluid into the line, contact the IV Access Service for advice.

Alteplase must be checked by 2 nurses, one of whom must be a Registered Nurse.

Administration is to be conducted by a Registered Nurse or IV Medication Competent Enrolled Nurse.

Dosage is determined by the lumen priming volume of the catheter. The lumen priming volume for an Infusaport is **2mls / lumen**.

- Unblocking a single lumen infusaport:

Administer 2mls of reconstituted Alteplase into the affected lumen.

- Unblocking a double lumen infusaport:
Administer 2mls of reconstituted Alteplase into each affected lumen.

Repeat administration of Alteplase within a 24-hour period if lumen remains sluggish.

6.3 Procedural Information

PART 1: PREPARATION AND INSTILLATION OF ALTEPLASE

Requirements

- Disposable kidney dish
- 2 x 10ml BD Posiflush™ Pre-Filled Saline Syringes or 2 x Sodium Chloride 0.9% ampoules for injection and 2 x 10 ml Luer Lock syringes
- 2 x large 70% alcohol wipes
- PPE including non-sterile gloves ^{6,8}
- 1 x 2mg/2mL Alteplase prefilled syringe
- 1 x Water for injection ampoule (for vial only)
- 1 x 10ml Luer Lock syringe for Alteplase (2 required if more than one lumen involved)
- 1 x drawing up needle (2 required if more than one lumen involved)

Procedure

1. Perform hand hygiene.⁶
2. Clean dressing trolley with combined detergent/disinfectant wipes.
3. Gather equipment for the procedure.
4. Perform hand hygiene.⁶
5. **Prefilled Alteplase 2mg/2mL syringe are supplied from Pharmacy**, visually inspect the contents of the syringe to confirm it has completely defrosted prior to administration.
 - a) The Alteplase is supplied reconstituted from pharmacy in a pre-filled syringe, decant the dose of Alteplase into a 10mL syringe and administer.
6. Perform hand hygiene and apply non-sterile gloves.
7. Scrub the lumen using a non-touch technique with a 70% alcohol wipe for 30 seconds using different areas of the wipe ⁶. Allow to air dry.
8. Flush with 0.9% Sodium Chloride for injection using a 10ml syringe, or BD Posiflush™ syringe.
9. Using non-touch technique, instil Alteplase 2ml slowly using a gentle pulsating action. Note this may take some time. Instil a small amount at a time.
10. Label the device with a sticker indicating Alteplase insitu, including date and time.⁷
11. Dispose of waste appropriately.
12. Remove gloves and perform hand hygiene.
13. Clean trolley with combined detergent/disinfectant wipes.
14. Perform hand hygiene.
15. Document date and time of instillation on the medication chart, and label the lumen to identify catheter lock insitu.
16. Allow to dwell for a minimum of 1 hour before attempting to aspirate. Ideally leave for 4 hours or, optimally, up to 24 hours.

PART 2: ASPIRATION OF ALTEPLASE

Requirements

- Disposable kidney dish
- PPE including non-sterile gloves

- Large 70% Alcohol wipe
- 2 x Sodium Chloride 0.9% ampoules and 2 x 10ml Luer Lock syringes and drawing up needle or 2 x BD Posiflush™ Pre-filled Saline Syringes^{10, 17, 18}
- 1 x 10ml Luer Lock syringe

Additional requirements for “open-ended” InfusaPorts

- 1 x 10ml Luer Lock syringe
- Heparinised saline 50 units in 5mls
- 1 x drawing up needle

Procedure

1. Repeat steps 1 - 4 above.
2. Apply non-sterile gloves and other PPE as indicated.
3. Scrub the lumen using a non-touch technique with a large 70% alcohol wipe for 30 seconds using different areas of the wipe⁶. Allow to air dry.
4. Attempt to aspirate Alteplase from lumen with 5ml syringe. A 5mL syringe is only to be used for Alteplase withdrawal. If unable to aspirate, the Alteplase may be safely flushed into the catheter.
5. Once the blockage is cleared, flush well with 10-20mls Sodium Chloride 0.9%.
 - **NOTE: For an open-ended InfusaPort only:** Follow Normal Saline flush with a flush of Heparinised Saline 50units in 5mls using a rapid pulsating push/pause action.
6. Discard of waste and remove gloves
7. Perform hand hygiene.^{1,7,8,11}
8. Document in the digital medical record.
9. If the catheter remains resistive to flush or aspirate blood post Alteplase administration, contact the IV Access Service for advice.

Related Documents

Legislation:

- Health Practitioner Regulation National Law (WA) Act 2010
- Carer's Recognition Act 2004 (WA).
- Work Health and Safety Act 2020
- Nursing and Midwifery Board of Australia (NMBA) Code of Professional Conduct for Nurses March 2018
- Western Australia: Medicines and Poisons Act 2014, Medicine and Poisons Regulations 2016, Pharmacy Act 2010

Policy:

- WA Health MP013/20 High Risk Medication Policy
- WA Health Consent to Treatment Policy
- NMHS Consent to Treatment Policy
- NMHS Clinical Documentation Policy
- SCGOPHCG Infection Prevention and Management Manual - Aseptic Technique Policy
- SCGOPHCG Transfusion Policy; Specimen Collection

Guideline

- SCGOPHCG Medication Management Guideline – Mupirocin Nasal Ointment for Decolonisation
- SCGOPHCG Nursing Practice Guideline: Intravenous Therapy

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